AN ORDINAL-PRIME TRANSFORM OF THE MÖBIUS FUNCTION

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1. Purpose and scope

This note develops the ordinal-prime transform of the Möbius function and determines exactly how much arithmetic information it carries. It is self-contained and uses no other construction.

The starting point is the Möbius function $\mu(n) \in \{-1, 0, +1\}$. The occurrences of -1 and of $+1$ are numbered separately, in increasing order of n, and the corresponding ordinal prime is attached to each occurrence; zeros are mapped to 1. The resulting signed sequence is $S(n)$ and its partial sum is $SS(n) = \sum_{k \le n} S(k)$.

The conclusions are:

(i) $SS(n)$ has an exact closed form in terms of the Mertens function $M(n)$, the squarefree counting function $Q(n)$, and the sum of the first m primes (Proposition 3.3).

Date: Version 4.

(ii) After subtracting the deterministic zero contribution $n - Q(n)$, the remainder $T(n) := SS(n) - n + Q(n)$ is *exactly* a signed sum of $|M(n)|$ consecutive primes, occupying a window of prime indices symmetric about $(Q(n)+1)/2$ (Proposition 4.1).

(iii) Consequently $T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}(1 + \theta(n))$ with $\theta(n) \to 0$ effectively, and the Mertens criterion transfers to T in both directions (Theorem 6.2).

(iv) Every object built here — SS, T, the residual E, and the centered residual E_{mid} — is a deterministic function of the pair $(M(n), Q(n))$ alone. The transform therefore creates no independent arithmetic constraint on $M(n)$. This applies to the residuals as well and is stated as Proposition 8.1.

(v) The nonlinearity of the transform splits exactly into a central part linear in $M(n)$ and an *antisymmetric weighted sum of prime gaps* with triangular weights (Theorem 10.1). The weight norm has a closed form (Lemma 10.2), which forces the exponent $3/2$ in the observed law $|E_{\mathrm{mid}}(n)| \approx 0.155\, |M(n)|^{3/2} \log n$.

(vi) The measured size of that residual is quantitatively accounted for by the local covariance structure of prime gaps. A covariance-corrected model predicts its standard deviation to within 0.3% with no free parameter, whereas an independent-gap model underpredicts by a factor 0.853 (Section 12, part D).

(vii) The packet structure visible in E_{mid} is an amplitude effect: its moving RMS follows $|M(n)|^{3/2} \log n$. On the logarithmic coordinate $u = \log n$, the spectrum of $M(e^u)e^{-u/2}$ displays the ordinates γ_k of zeta zeros, as predicted by the classical explicit formula. This spectral observation is a sanity check, not new information about the zeros; the finite u-window has frequency resolution about 0.91.

(viii) A separate test asks whether the prime-gap fluctuation factor is coupled to the sign or zeta phase of M. No such coupling is detected at the present scale. With effective sample about 1.7×10^4, the tests only exclude linear effects larger than roughly 0.015 and do not constitute a theorem of independence.

All numerical statements in this note were verified for $1 \le n \le 10^6$; the verification protocol and effective ranges are given in Section 12.

2. Notation and elementary lemmas

Let $p_1 = 2, p_2 = 3, p_3 = 5, \ldots$ denote the primes in increasing order, and set $p_0 := 0$ by convention (used only to make the product forms below well defined at $n = 1$).

Write

$$M(n) := \sum_{k \le n} \mu(k), \qquad Q(n) := \sum_{k \le n} \mu(k)^2, \qquad \mathcal{P}(m) := \sum_{j=1}^{m} p_j, \quad \mathcal{P}(0) := 0.$$

Define the cumulative occurrence counts

$$N_-(n) = \sum_{k \le n} \frac{\mu(k)^2 - \mu(k)}{2}, \qquad N_0(n) = \sum_{k \le n} \big(1 - \mu(k)^2\big), \qquad N_+(n) = \sum_{k \le n} \frac{\mu(k)^2 + \mu(k)}{2}.$$

Thus $N_-(n)$, $N_0(n)$, $N_+(n)$ count the occurrences of $\mu = -1$, $\mu = 0$, $\mu = +1$ up to n, and the counts are inclusive of n itself.

Lemma 2.1 (Counter identities)**.** *For every $n \ge 1$,*

$$N_+(n) + N_-(n) = Q(n), \qquad N_+(n) - N_-(n) = M(n), \qquad N_0(n) = n - Q(n),$$

hence

$$N_+(n) = \frac{Q(n) + M(n)}{2}, \qquad N_-(n) = \frac{Q(n) - M(n)}{2}.$$

Lemma 2.2 (Parity)**.** $Q(n) \equiv M(n) \pmod 2$ *for every $n \ge 1$. Consequently the indices $(Q(n) \pm M(n))/2$ occurring throughout are integers.*

Proof. $Q = N_+ + N_-$ and $M = N_+ - N_-$, so $Q - M = 2N_-$. □

Lemma 2.2 is used silently in every subsequent formula and was absent from version 1; without it the central identities are not well formed.

3. The transform

Definition 3.1. The three rows are

$$R_-(n) = -\frac{\mu(n)^2 - \mu(n)}{2}\, p_{N_-(n)}, \qquad R_0(n) = 1 - \mu(n)^2, \qquad R_+(n) = \frac{\mu(n)^2 + \mu(n)}{2}\, p_{N_+(n)},$$

with the convention $p_0 = 0$ (needed only at $n = 1$, where $N_-(1) = 0$ and the coefficient vanishes). The ordinal-prime transform is

$$S(n) := R_-(n) + R_0(n) + R_+(n) = \begin{cases} -p_{N_-(n)}, & \mu(n) = -1, \\ 1, & \mu(n) = 0, \\ +p_{N_+(n)}, & \mu(n) = +1. \end{cases}$$

Along the occurrences of $\mu = -1$ the values are $-2, -3, -5, -7, -11, \ldots$; along the occurrences of $\mu = +1$ they are $2, 3, 5, 7, 11, \ldots$, in a separate ordinal count. Explicitly,

$$S(1), \ldots, S(12) = 2,\ -2,\ -3,\ 1,\ -5,\ 3,\ -7,\ 1,\ 1,\ 5,\ -11,\ 1.$$

Definition 3.2. $SS(n) := \sum_{k=1}^{n} S(k)$ and $T(n) := SS(n) - \big(n - Q(n)\big)$.

Proposition 3.3 (Closed form)**.** *For every $n \geq 1$,*

$$\boxed{\begin{aligned} SS(n) &= \mathcal{P}\big(N_+(n)\big) - \mathcal{P}\big(N_-(n)\big) + N_0(n) \\ &= \mathcal{P}\left(\frac{Q(n) + M(n)}{2}\right) - \mathcal{P}\left(\frac{Q(n) - M(n)}{2}\right) + n - Q(n). \end{aligned}}$$

Proof. The $k \leq n$ with $\mu(k) = +1$ receive, in order, the primes $p_1, \ldots, p_{N_+(n)}$, contributing $\mathcal{P}(N_+(n))$; those with $\mu(k) = -1$ contribute $-\mathcal{P}(N_-(n))$; each of the $N_0(n)$ zeros contributes 1. The second form is Lemma 2.1, the indices being integers by Lemma 2.2. □

4. The signed prime block

Proposition 4.1 (Block representation)**.** *Write $W(n)$ for the set of prime indices strictly between $N_-(n)$ and $N_+(n)$, inclusive of the larger endpoint; $W(n)$ has exactly $|M(n)|$ elements and is symmetric about the index $(Q(n) + 1)/2$. Then*

$$T(n) = \operatorname{sgn}\big(M(n)\big) \sum_{j \in W(n)} p_j = \begin{cases} \displaystyle\sum_{j=(Q-M)/2+1}^{(Q+M)/2} p_j, & M(n) > 0, \\ 0, & M(n) = 0, \\ \displaystyle -\sum_{j=(Q-|M|)/2+1}^{(Q+|M|)/2} p_j, & M(n) < 0. \end{cases}$$

In particular

$$\operatorname{sgn} T(n) = \operatorname{sgn} M(n), \qquad T(n) = 0 \iff M(n) = 0.$$

Proof. Immediate from Proposition 3.3: $T = \mathcal{P}(N_+) - \mathcal{P}(N_-)$ is a telescoping difference of prefix sums of the primes. The window endpoints are $\min(N_\pm) + 1$ and $\max(N_\pm)$, whose midpoint is $(N_+ + N_- + 1)/2 = (Q + 1)/2$. □

Corollary 4.2 (Exact bounds)**.**

$$|M(n)|\, p_{\frac{Q(n) - |M(n)|}{2} + 1} \;\leq\; |T(n)| \;\leq\; |M(n)|\, p_{\frac{Q(n) + |M(n)|}{2}}.$$

Both inequalities are trivial (each of the $|M(n)|$ primes in the block lies between the smallest and the largest one), but the *lower* bound is what makes the transfer of Mertens bounds an equivalence rather than an implication; see Theorem 6.2.

5. The prime scale, with explicit error

We record the multiplicative estimate in a form that is valid at every n, including the infinitely many n with $M(n) = 0$. (Version 1 wrote $T(n) \sim M(n)p_{Q(n)/2}$; that assertion is empty on the zero set of M, which contains 5361 values of $n \le 10^6$.)

Proposition 5.1. *Define $\theta(n) := 0$ if $M(n) = 0$ and otherwise by $T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}\big(1 + \theta(n)\big)$. Then*

$$|\theta(n)| \;\le\; \max_{|j - Q(n)/2| \le |M(n)|/2+1} \left| \frac{p_j}{p_{\lfloor Q(n)/2 \rfloor}} - 1 \right|,$$

and $\theta(n) \to 0$ as $n \to \infty$.

Proof. The bound is Proposition 4.1 together with the triangle inequality. For the limit: by the prime number theorem in the form $M(n) = o(n)$, and by $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$, one has $|M(n)|/Q(n) \to 0$; hence the window endpoints are $(1 + o(1))Q(n)/2$ on the index scale, and $p_{(1+o(1))m}/p_m \to 1$ by the prime number theorem again. □

Both hypotheses invoked here are named: $M(n) = o(n)$ is equivalent to the prime number theorem, and $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$ is elementary.

Remark 5.2 (The asymptotic constant is a poor numerical guide)**.** From $p_m = m(\log m + \log\log m - 1 + o(1))$ and $Q(n)/2 \sim \frac{3}{\pi^2} n$ one gets

$$p_{\lfloor Q(n)/2 \rfloor} = \frac{3}{\pi^2}\, n \log n \times \left(1 + \frac{\log\log n - 1 + \log(3/\pi^2)}{\log n} + o\Big(\tfrac{1}{\log n}\Big)\right),$$

so the correction to the leading constant decays only like $1/\log n$. Measured values are

$$\frac{T(n)}{(3/\pi^2) n \log n\, M(n)} = 0.9558,\ 0.9954,\ 1.0151,\ 1.0281$$

at $n = 10^3, 10^4, 10^5, 10^6$; they are not monotone and remain 2.8% off at $n = 10^6$. By contrast,

$$\frac{T(n)}{M(n) p_{\lfloor Q(n)/2 \rfloor}} = 1.00200,\ 1.00071,\ 1.000097,\ 0.999998$$

at the same points. Any quantitative use should keep $p_{\lfloor Q(n)/2 \rfloor}$ and not the closed constant.

6. Transfer of the Mertens criterion

Proposition 6.1. *$|T(n)| = O\big(n \log n\, |M(n)|\big)$ and hence $SS(n) = O\big(n \log n\, |M(n)| + n\big)$. Unconditionally $SS(n) = o(n^2 \log n)$, by $M(n) = o(n)$ (prime number theorem).*

Theorem 6.2. *The following are equivalent.*

(a) The Riemann hypothesis.
(b) $M(n) = O_\varepsilon(n^{1/2+\varepsilon})$ for every $\varepsilon > 0$.
(c) $SS(n) - n + Q(n) = O_\varepsilon(n^{3/2+\varepsilon})$ for every $\varepsilon > 0$.

Proof. (a) $\iff$ (b) is classical (Littlewood).

(b)⇒(c): by Corollary 4.2, $|T(n)| \le |M(n)| p_{(Q+|M|)/2} \ll |M(n)|\, n \log n \ll_\varepsilon n^{3/2+\varepsilon} \log n$, and the logarithm is absorbed into n^ε.

(c)⇒(b): this is the direction version 1 asserted without proof. By the prime number theorem $|M(n)| = o(n)$ and $Q(n) = \frac{6}{\pi^2} n + O(\sqrt{n})$, so

$$\frac{Q(n) - |M(n)|}{2} + 1 = \big(1 + o(1)\big) \frac{3}{\pi^2} n,$$

whence the smallest prime of the block satisfies $p_{(Q-|M|)/2+1} \geq c\, n \log n$ for some $c > 0$ and all large n. The lower bound of Corollary 4.2 then gives

$$|M(n)| \leq \frac{|T(n)|}{c\, n \log n} \ll_\varepsilon \frac{n^{3/2+\varepsilon}}{n \log n} \ll_\varepsilon n^{1/2+\varepsilon}. \qquad \square$$

Remark 6.3. Theorem 6.2 is a reformulation, not a proof mechanism. Note also that (b) must not be strengthened to $M(n) = O(\sqrt{n})$: the Riemann hypothesis does not imply the Mertens conjecture, which is false (Odlyzko–te Riele, 1985).

7. The zeta ingredients

The two Möbius-derived ingredients have the classical Dirichlet series

$$\sum_{m\geq 1} \frac{\mu(m)}{m^s} = \frac{1}{\zeta(s)}, \qquad \sum_{m\geq 1} \frac{\mu(m)^2}{m^s} = \frac{\zeta(s)}{\zeta(2s)}, \qquad \Re s > 1,$$

and the prime weights are governed by $\log \zeta(s)$ through the prime-counting function, via the exact Abel identity

$$\mathcal{P}(m) = m\, p_m - \int_2^{p_m} \pi(t)\, dt,$$

(verified here for $m \leq 100$; e.g. $\mathcal{P}(3) = 10 = 3 \cdot 5 - 5$).

Version 1 substituted Perron integrals for M and Q into the arguments of $\mathcal{P}$. That is not a functional identity and it is dropped here, for two reasons. First, $\mathcal{P}$ is defined only on nonnegative integers, so an integral placed inside its argument is notation for "the integer that this integral evaluates to", not an analytic representation. Second, at integer x the untruncated Perron integral returns $M(x) - \frac{1}{2}\mu(x)$, not $M(x)$.

The honest statement is the following. The transform composes three classical objects,

$$\frac{1}{\zeta(s)} \ \text{(signed Möbius imbalance)}, \qquad \frac{\zeta(s)}{\zeta(2s)} \ \text{(squarefree count)}, \qquad \log \zeta(s) \ \text{(prime weights)},$$

through the nonlinear expression $\mathcal{P}\big(\frac{Q+M}{2}\big) - \mathcal{P}\big(\frac{Q-M}{2}\big) + n - Q$. None of the three components is new, and the composition is not a Dirichlet series of anything.

8. Exact information content

Proposition 8.1. *For each n, $SS(n)$ is a deterministic function of the pair $(M(n), Q(n))$; and with $Q(n)$ held fixed, $T(n)$ is strictly increasing in $M(n)$. The same is true of every residual defined below, including $E(n)$ and $E_{\mathrm{mid}}(n)$.*

Proof. The first claim is Lemma 2.1 plus Proposition 3.3. For the second, increasing M by 2 at fixed Q raises N_+ by one and lowers N_- by one, changing T by $p_{N_++1} + p_{N_-} > 0$. The residuals E and E_{mid} are T minus an explicit function of (M, Q), hence also functions of (M, Q). $\square$

Two consequences deserve to be stated plainly.

First, $SS(n)$ does not retain the internal ordering of the ± 1 values of μ below n: two Möbius sign histories with the same endpoint counts produce the same $SS(n)$. It is a nonlinear endpoint statistic.

Second — and this corrects a claim in version 1 — the residual $E(n)$ is *not* structurally distinct from $SS(n)$ in the informational sense. Its window is determined by $(M(n), Q(n))$, so the argument above applies to it verbatim. What is genuinely different about the residual is not its information content but its *size*, which turns out to be governed by prime-gap statistics; that is the subject of the next two sections.

There are two distinct meanings of "information" that must not be mixed. In the logical sense, Proposition 8.1 is final: $E_{\mathrm{mid}}(n) = G(M(n), Q(n))$ for an explicit deterministic G, so no experiment can make E_{mid} contain arithmetic information absent from the endpoint pair. In the analytic or *leverage* sense, one can still ask whether a separately proved bound for the prime-gap fluctuation factor could combine with the identity to constrain M. A bound for T

itself cannot do this: at fixed Q, T is strictly monotone in M and the two-sided block bounds make a Mertens bound and the corresponding T bound equivalent up to the explicit prime scale. What remains testable is only whether some independently controlled feature of the gap fluctuation is coupled to the sign, phase, or trajectory of M. Section 12 measures a small family of such couplings and finds none at the present resolution; this is a negative numerical result, not a logical strengthening of Proposition 8.1.

9. The residual and its correct centering

Definition 9.1. Let

$$p^*(n) := \tfrac{1}{2}\Big(p_{\lfloor (Q(n)+1)/2 \rfloor} + p_{\lceil (Q(n)+1)/2 \rceil}\Big)$$

be the linear interpolation of the prime sequence at the exact window centre $(Q(n)+1)/2$ of Proposition 4.1. Define

$$E(n) := T(n) - M(n)\, p_{\lfloor Q(n)/2 \rfloor}, \qquad E_{\mathrm{mid}}(n) := T(n) - M(n)\, p^*(n),$$

so that the exact decomposition

$$\boxed{T(n) = M(n)\, p^*(n) + E_{\mathrm{mid}}(n)}$$

holds for every n.

When $Q(n)$ is odd, $(Q+1)/2$ is an integer and $p^* = p_{(Q+1)/2}$; when $Q(n)$ is even, p^* is the mean of the two central primes.

Proposition 9.2 (Centering offset)**.** *$E(n) - E_{\mathrm{mid}}(n) = M(n)\big(p^*(n) - p_{\lfloor Q(n)/2 \rfloor}\big)$, where the bracket equals one half of a prime gap when $Q(n)$ is even and one full prime gap when $Q(n)$ is odd. It is positive and of size $\asymp \log p^*(n)$, so E carries a deterministic component linear in $M(n)$ which E_{mid} does not.*

The identity was verified exactly for all $n \le 10^6$; the median of $\big(p^* - p_{\lfloor Q/2 \rfloor}\big)/\log p_{\lfloor Q/2 \rfloor}$ over $10^4 \le n \le 10^6$ is 0.5318. Proposition 9.2 explains the correlation structure of the two residuals: the offset is a fixed positive multiple of $M(n)$, so E correlates with M and E_{mid} does not. Measured over $100 \le n \le 10^6$,

$$\mathrm{corr}(E, M) = +0.2573, \qquad \mathrm{corr}(E_{\mathrm{mid}}, M) = +0.0045.$$

E_{mid} is therefore the object to study; E is E_{mid} plus a bookkeeping artefact. Version 1 studied only E.

10. Exact antisymmetric gap representation

Write $g_i := p_{i+1} - p_i$ and $m(n) := \lfloor (Q(n)+1)/2 \rfloor$.

Theorem 10.1 (Gap representation)**.** *Let $h := |M(n)|$. Then*

$$\boxed{E_{\mathrm{mid}}(n) = \mathrm{sgn}\big(M(n)\big) \sum_i w_i\, g_i}$$

where the weights w_i are supported on $h-1$ or $h-2$ gaps symmetrically placed about $m = m(n)$, are antisymmetric in sign about the centre, and are given explicitly by

$$h \text{ even}: \quad w_{m+k} = \tfrac{h}{2} - k, \quad w_{m-k} = -\big(\tfrac{h}{2} - k\big), \quad 1 \le k \le \tfrac{h}{2} - 1;$$

$$h \text{ odd}: \quad w_{m+k} = \tfrac{h-1}{2} - k \ \ (0 \le k \le \tfrac{h-3}{2}), \qquad w_{m-k} = -\big(\tfrac{h+1}{2} - k\big) \ \ (1 \le k \le \tfrac{h-1}{2}).$$

In both cases the gap at the centre cancels identically, and the multiset of weight magnitudes on each side is $\{1, 2, \ldots, \lfloor (h-1)/2 \rfloor\}$.

Proof. Take $M(n) > 0$; the other sign is identical after a global change of sign.

Suppose h is even, so Q is even by Lemma 2.2, $m = Q/2$ and $p^* = \frac{1}{2}(p_m + p_{m+1})$. The window is $\{m - \frac{h}{2} + 1, \dots, m + \frac{h}{2}\}$. For $j > m$, $p_j - p^* = \frac{1}{2}g_m + \sum_{i=m+1}^{j-1} g_i$; for $j \le m$, $p_j - p^* = -\frac{1}{2}g_m - \sum_{i=j}^{m-1} g_i$. Summing over the window, the $\frac{1}{2}g_m$ terms occur $\frac{h}{2}$ times with each sign and cancel. Counting how many j contain a given g_{m+k} gives the coefficient $\frac{h}{2} - k$, and symmetrically $-(\frac{h}{2} - k)$ for g_{m-k}.

If h is odd then Q is odd, $m = (Q+1)/2$ and $p^* = p_m$, the window being $\{m - \frac{h-1}{2}, \dots, m + \frac{h-1}{2}\}$. The same count gives the stated coefficients, the two sides again carrying magnitudes $\frac{h-1}{2}, \dots, 1$. □

Lemma 10.2 (Weight norm)**.**

$$\boxed{\sum_i w_i^2 = \begin{cases} \dfrac{h(h^2-1)}{12}, & h \text{ odd}, \\ \dfrac{h(h-1)(h-2)}{12}, & h \text{ even}, \end{cases} \qquad \text{so} \qquad \sum_i w_i^2 = \frac{h^3}{12}\big(1 + O(h^{-1})\big).}$$

Proof. By Theorem 10.1 each side contributes $\sum_{j=1}^{\lfloor (h-1)/2 \rfloor} j^2$. For h odd this is $2 \cdot \frac{(\frac{h-1}{2})(\frac{h+1}{2})h}{6} = \frac{h(h^2-1)}{12}$; for h even it is $2 \cdot \frac{(\frac{h}{2}-1)(\frac{h}{2})(h-1)}{6} = \frac{h(h-1)(h-2)}{12}$. □

Theorem 10.1 and Lemma 10.2 are exact and unconditional. They are the reason the exponent $3/2$ appears below: it is forced by the triangular weights, not fitted.

Remark 10.3. Theorem 10.1 does *not* evade Proposition 8.1: the window position $m(n)$ and its length h are determined by $(M(n), Q(n))$, and the primes are deterministic, so E_{mid} remains a function of $(M(n), Q(n))$. What the theorem changes is what the *size* of E_{mid} is controlled by, namely the local second-order statistics of prime gaps.

Proposition 10.4 (Prime-counting integral form)**.** *Let $h = |M(n)|$, $a = \min(N_-(n), N_+(n))$ and $b = \max(N_-(n), N_+(n)) = a + h$. Then, for $h > 0$,*

$$\boxed{E_{\text{mid}}(n) = \operatorname{sgn}(M(n)) \left\{ \int_{p_a}^{p_b} \big(b - \pi(t)\big)\, dt - h\big(p^*(n) - p_a\big) \right\}.}$$

For $h = 0$ both sides are 0 by convention.

Proof. On each interval $[p_j, p_{j+1})$, $\pi(t) = j$, hence

$$\int_{p_a}^{p_b} (b - \pi(t))\, dt = \sum_{j=a}^{b-1} (b-j)(p_{j+1} - p_j) = \sum_{j=a+1}^{b} (p_j - p_a).$$

The last expression is the unsigned prime block minus hp_a. Subtracting $h(p^* - p_a)$ gives the unsigned block minus hp^*, and the global factor $\operatorname{sgn}(M)$ gives E_{mid}. □

Thus each value of E_{mid} may also be read as an integrated prime counting discrepancy over the short interval $[p_a, p_b]$, whose length is of order $h \log p_a$. This is genuine information about prime gaps in that specific window, even though the choice of the window itself is already fixed by (M, Q) and hence cannot create a new restriction on M.

11. Two variance models

Theorem 10.1 suggests modelling E_{mid} as a weighted sum of gap fluctuations. Two models are compared; both are heuristics, and both were tested against the data of Section 12.

Definition 11.1 (Independent-gap model)**.** Treat the g_i near index $m(n)$ as uncorrelated with common local variance $\sigma_g(n)^2$. Then by Lemma 10.2

$$\mathrm{SD}_{\text{ind}}(n) := \sigma_g(n) \sqrt{\textstyle\sum_i w_i^2} \;\approx\; \frac{\sigma_g(n)}{\sqrt{12}}\, h^{3/2}.$$

Definition 11.2 (Covariance-corrected model)**.** Let $\rho(d)$ be the local autocorrelation of the gap sequence, and put $R_{ab} = \rho(|i_a - i_b|)$ on the support of w. Define

$$\kappa(h) := \frac{w^\top R w}{w^\top w}, \qquad \mathrm{SD}_{\mathrm{cov}}(n) := \sigma_g(n)\sqrt{\kappa(h)\sum_i w_i^2}.$$

For fixed h the correction factor contains no statistic beyond the gap autocorrelation itself. If

$$c_d(h) := \sum_{|i-j|=d} w_i w_j, \qquad c_0(h) = \sum_i w_i^2,$$

then exactly

$$\boxed{\kappa(h) = \frac{1}{c_0(h)} \sum_{d \geq 0} c_d(h)\rho(d).} \tag{1}$$

Thus κ is the autocorrelation sequence viewed through a family of triangular antisymmetric kernels. Numerically, $\max|\kappa_{\mathrm{direct}} - (C\rho)/c_0| = 7 \times 10^{-8}$. Solving the finite linear system in the reverse direction reproduces the first gap correlations

$$-0.0463,\ -0.0225,\ -0.0132,\ -0.0057,\ -0.0069,\ -0.0061,\ -0.0060,\ -0.0064.$$

The maximum error reported for $d \leq 60$ is 6.8×10^{-4}. The inversion is severely ill-conditioned (condition number about 10^8), so in practice only the first lags are stably recoverable. This reinforces the information statement: $\kappa(h)$ is a compressed linear image of $\rho(d)$, not an additional spectral observable.

Because w is antisymmetric about the centre, a *positive* correlation between gaps on opposite sides of the centre enters $w^\top R w$ with a negative sign. One therefore expects $\kappa(h) < 1$, i.e. the independent model to *overpredict*. This is what is observed, and it accounts quantitatively for the discrepancy (Section 12, part D).

12. Numerical verification and measured laws

Protocol. μ was computed by a linear sieve for $n \leq N = 10^6$; M, Q by prefix sums; a prime table to index 304 071 ($p = 4\,318\,813$); T by prefix sums of the primes. At $N = 10^6$: $M(N) = 212$, $Q(N) = 607\,926$, maximum required prime index 304 069. All scripts and raw outputs accompany this note.

A. Exact statements verified. The following hold with no exceptions for all $1 \leq n \leq 10^6$: Lemma 2.2; Proposition 3.3 (direct summation of S against the closed form); Proposition 4.1 and both inequalities of Corollary 4.2; $\operatorname{sgn} T = \operatorname{sgn} M$ and $T = 0 \iff M = 0$ (there are 5 361 values of $n \leq 10^6$ with $M(n) = 0$); Proposition 9.2. Theorem 10.1 was checked on 4 000 random $n \in [10^3, 10^6]$ with maximum absolute error 0, and Lemma 10.2 for all $h \leq 60$ and $h \in \{100, 101, 300, 301, 370\}$.

B. Global residual scale, $100 \leq n \leq 10^6$**.**

	E	E_{mid}
median $\lvert\cdot\rvert$	1.09×10^3	9.74×10^2
r.m.s.	5.295×10^3	5.066×10^3
max $\lvert\cdot\rvert$	6.2766×10^4	5.7654×10^4
median $\lvert\cdot\rvert/\lvert T\rvert$	1.069×10^{-5}	8.894×10^{-6}

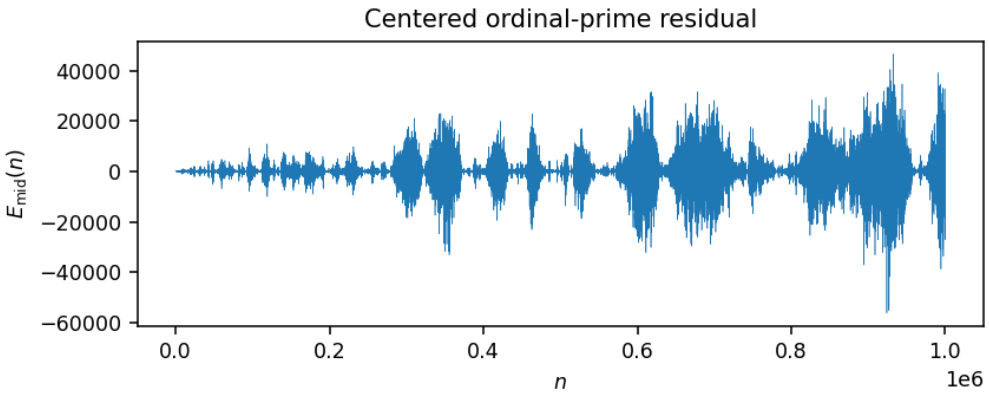


FIGURE 1. $E_{\mathrm{mid}}(n)$ for $n \le 10^6$.

C. The $h^{3/2}$ law and its stationarity. Put $Z_{3/2}(n) := E_{\mathrm{mid}}(n)/\big(|M(n)|^{3/2}\log n\big)$. Conditioning on $h = |M(n)|$ over $n \ge 10^3$:

$h \in$	$[8,12)$	$[12,19)$	$[19,31)$	$[31,51)$	$[51,84)$	$[84,138)$	$[138,225)$	$[225,400)$
median $\lvert Z_{3/2}\rvert$	0.1426	0.1477	0.1589	0.1589	0.1556	0.1546	0.1507	0.1550
count	36 451	58 125	86 041	127 696	152 059	164 880	231 986	65 791

and conditioning on n instead:

$n \in$	RMS $Z_{3/2}$	median $\lvert Z_{3/2}\rvert$	count
$[1.00, 1.58) \times 10^5$	0.2331	0.1549	51 624
$[1.58, 2.51) \times 10^5$	0.2379	0.1574	82 660
$[2.51, 3.98) \times 10^5$	0.2340	0.1541	134 286
$[3.98, 6.31) \times 10^5$	0.2314	0.1530	215 541
$[6.31, 10.0) \times 10^5$	0.2324	0.1552	359 873

Both tables are flat: the normalisation $|M|^{3/2}\log n$ is stable across a factor of 46 in h and a factor of 10 in n. This supports, but does not prove, the hypothesis that $Z_{3/2}$ converges in distribution.

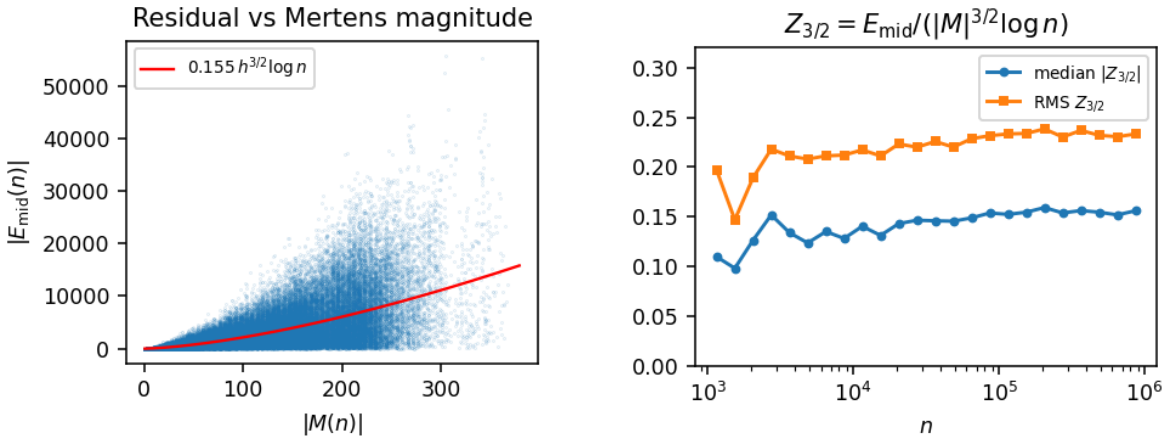


FIGURE 2. Left: $|E_{\mathrm{mid}}|$ against $|M|$ with the envelope $0.155\,h^{3/2}\log n$. Right: stationarity of $Z_{3/2}$ in n.

D. Which variance model is right. Local gap statistics were measured on the prime-index band $[5 \times 10^4, 3.05 \times 10^5)$, which covers the windows for $n \in [10^5, 10^6]$: mean gap 14.5927, $\log p = 14.6984$ at the band centre, $\mathrm{Var}(g) = 145.93$, hence

$$\mathrm{Var}(g)/\log^2 p = 0.6755$$

against the value 1 predicted by a Poisson (Cramér) gap model. The gap autocorrelation is negative at short lag and positive over a long tail:

d	1	2	3	4	5	6
$\rho(d)$	−0.0463	−0.0225	−0.0132	−0.0057	−0.0069	−0.0061

with $\sum_{1 \le d \le 420} \rho(d) = +0.4646$. The resulting correction factor decreases steadily,

h	5	10	20	40	80	160	320
$\kappa(h)$	1.0206	0.9532	0.9094	0.8514	0.7869	0.7381	0.6871

Testing both models on the 883 123 values of n with $n \ge 10^4$ and $h \ge 12$:

model	$\mathrm{RMS}(E_{\mathrm{mid}})/\mathrm{RMS}(\mathrm{SD})$	RMS Z	mean Z	median $\lvert Z\rvert$
independent gaps	0.8527	0.8963	-1.88×10^{-4}	0.5962
covariance-corrected	1.0031	1.0036	-2.13×10^{-4}	0.6694

The 15% deficit of the independent model is therefore explained, to within 0.3%, by the correlations between prime gaps; and the corrected model reproduces the Gaussian value median $|Z| = 0.6745$ to within 0.8% with no free parameter. This settles the question left open by the independent model.

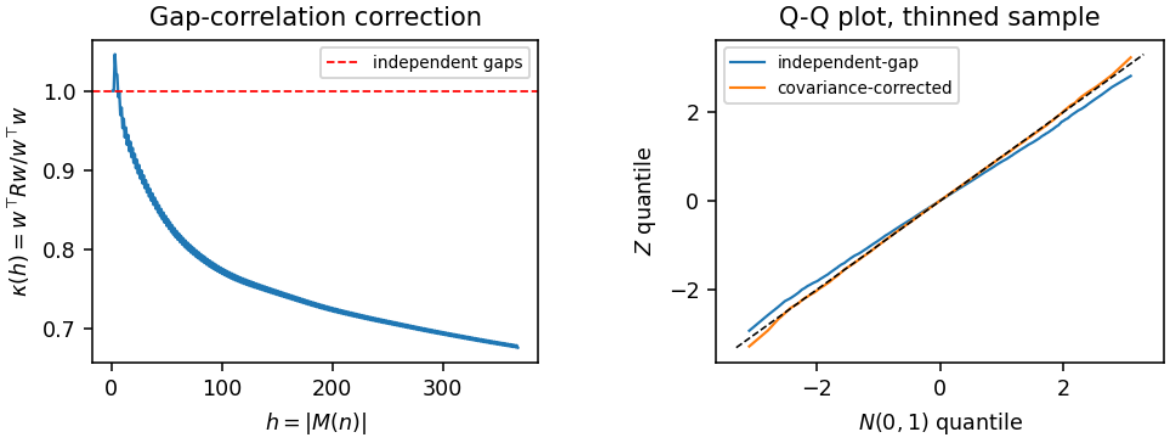


FIGURE 3. Left: the correction factor $\kappa(h)$. Right: quantiles of $Z = E_{\mathrm{mid}}/\mathrm{SD}$ against the normal, on a thinned sample; the independent-gap curve is visibly compressed, the corrected one is not.

E. Distribution of the standardised residual. With $Z := E_{\mathrm{mid}}/\mathrm{SD}_{\mathrm{cov}}$, by blocks in n ($h \ge 12$):

$n \in$	count	mean	sd	skew	exc. kurt.	median $\lvert Z\rvert$
$[10^4, 10^5)$	67 120	+0.0043	0.9813	−0.0098	+0.2556	0.6474
$[10^5, 3.16 \times 10^5)$	183 261	−0.0000	1.0029	−0.0167	+0.1559	0.6705
$[3.16 \times 10^5, 10^6)$	632 741	−0.0008	1.0061	−0.0188	+0.0957	0.6715

and by blocks in h, the excess kurtosis decreases monotonically,

$h \in$	$[12, 19)$	$[19, 31)$	$[31, 51)$	$[51, 84)$	$[84, 138)$	$[138, 225)$	$[225, 400)$
exc. kurt.	$+0.676$	$+0.338$	$+0.213$	$+0.031$	$+0.058$	$+0.024$	-0.143

which is the signature expected if a central limit theorem governs the sum of Theorem 10.1 as $h \to \infty$. Quantiles of Z over all $n \geq 10^4$, $h \geq 12$:

level	1%	5%	25%	50%	75%	95%	99%
observed	-2.372	-1.652	-0.666	$+0.000$	$+0.673$	$+1.641$	$+2.346$
$N(0, 1)$	-2.326	-1.645	-0.674	0	$+0.674$	$+1.645$	$+2.326$

F. The points are not independent. This is a necessary caveat on the preceding tables. Consecutive n share almost the same window, so the 883 123 values of Z are strongly dependent: the autocorrelation of Z in n is $+0.979$ at lag 1, $+0.692$ at lag 10, and first falls below 0.05 at lag 128. The integrated autocorrelation time is ≈ 51, giving an effective sample size of about 1.7×10^4, not 8.8×10^5.

Consequently a Kolmogorov–Smirnov test on the full sample is meaningless. On thinned subsamples:

thinning	count	sd	exc. kurt.	KS	p
1	883 123	1.0036	$+0.120$	0.00317	3.8×10^{-8}
51	17 317	1.0036	$+0.170$	0.00451	0.87
128	6 900	1.0057	$+0.018$	0.00775	0.80

At the effective sample size, normality of Z is not rejected. The apparent rejection on the full sample is an artefact of dependence, not evidence against the model.

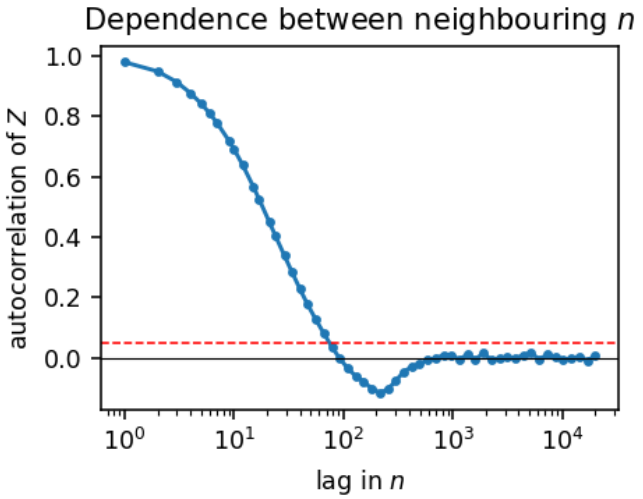


FIGURE 4. Autocorrelation of Z in n; the effective sample is roughly 50 times smaller than the nominal one.

G. Relative size of the residual. Combining $|E_{\mathrm{mid}}| \approx c\, h^{3/2} \log n$ with $|T| \approx h\, p^* \approx \frac{3}{\pi^2} h\, n \log n$ predicts

$$\frac{|E_{\mathrm{mid}}(n)|}{|T(n)|} \approx \frac{c\pi^2}{3} \cdot \frac{\sqrt{|M(n)|}}{n},$$

$$\frac{c\pi^2}{3} = 0.510 \qquad (c = 0.155).$$

Measured medians of $\big(|E_{\mathrm{mid}}|/|T|\big)\, n/\sqrt{|M|}$: 0.4385, 0.4800, 0.4995, 0.4955 on

$$[10^3, 10^4), \quad [10^4, 10^5), \quad [10^5, 3.16 \times 10^5), \quad [3.16 \times 10^5, 10^6).$$

This quantitatively explains why the central term reproduces T so accurately. The measured median satisfies

$$\text{median}\, \frac{|E_{\mathrm{mid}}|}{|T|} = 8.9 \times 10^{-6}.$$

If $|M(n)|$ is of size roughly $n^{1/2}$, the relative correction is of size $n^{-3/4}$.

H. Largest excursions. The extremes of E_{mid} on $n \le 10^6$ occur in tight clusters, e.g. -5.7654×10^4 at $n = 924\,067$ ($M = -301$) and $+4.6576 \times 10^4$ at $n = 932\,737$ ($M = -269$). These are the n at which $|M|$ is locally largest, consistent with the $h^{3/2}$ law and carrying no separate endpoint information.

I. The packet envelope. The visible packets are almost entirely the excursions of $|M(n)|$ passed through the $h^{3/2}$ variance scale. With a moving RMS window of width 2000, the log–log correlation between the measured envelope of E_{mid} and a moving average of $|M(n)|^{3/2} \log n$ is

$$0.9939 \qquad (n \ge 5 \times 10^4).$$

Replacing the Mertens factor by a trend depending only on n gives correlation 0.4442. Figure 5 displays the resulting amplitude modulation.

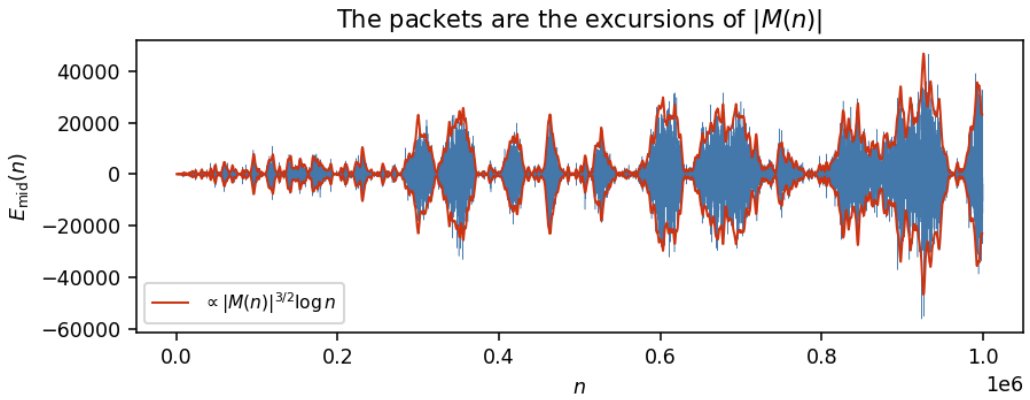


FIGURE 5. The packet structure of E_{mid} follows the excursions of $|M(n)|$. The red curves show a scaled moving average proportional to $|M(n)|^{3/2} \log n$.

J. Spectrum on the logarithmic coordinate: a classical check. Put $u = \log n$ and

$$X(u) := e^{-u/2} M(e^u).$$

The classical explicit formula for the Mertens function has, formally and after the usual truncations and qualifications, terms of the form

$$\frac{x^\rho}{\rho \zeta'(\rho)}.$$

Under RH, $\rho = \frac{1}{2} + i\gamma$ turns this after $x = e^u$ and multiplication by $e^{-u/2}$ into modes $e^{i\gamma u}$. Consequently the appearance of power at frequencies γ_k in $X(u)$ is the expected explicit-formula behaviour, not a new law of the transform.

The numerical spectrum over $10^3 \le n \le 10^6$ shows peaks near the first zeta ordinates, while a random-sign surrogate with the same squarefree support does not show the same visual alignment. The original peak-to-nearest-zero summary (mean distances 0.263 for the true sequence and 1.738 for one surrogate) has a selection bias because peaks were selected before matching

them to the zeros. It is retained only as a diagnostic and not used as evidence. The proper fixed-frequency question is the power evaluated at preassigned γ_k, or a regression on the preassigned modes.

More fundamentally, the finite logarithmic window itself imposes the resolution

$$\Delta\omega \simeq \frac{2\pi}{u_{\max} - u_{\min}} = \frac{2\pi}{\log(10^6/10^3)} \simeq 0.91.$$

Errors of a few tenths are therefore fully compatible with the known γ_k and carry no precision information about them. Doubling the frequency resolution requires doubling the range in u, hence squaring the range in n; reaching resolution 10^{-3} would require a logarithmic span of order $2\pi \times 10^3$, far outside this computation.

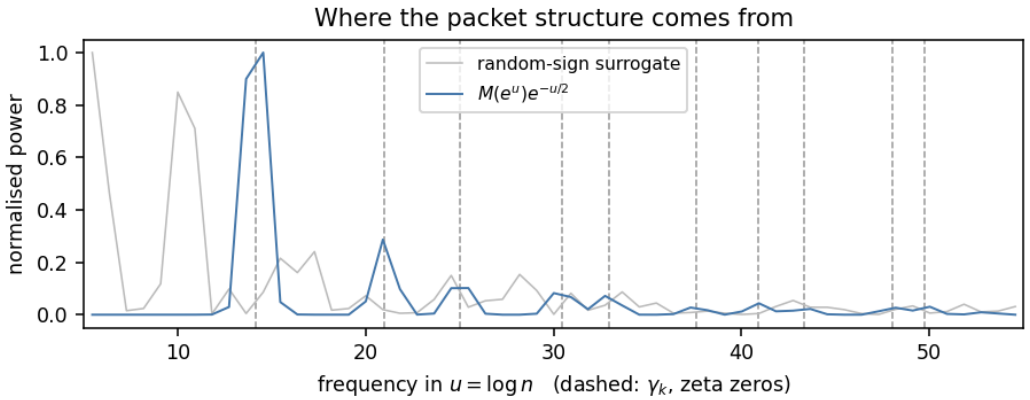


FIGURE 6. Power spectrum of $M(e^u)e^{-u/2}$ on $u = \log n$. Dashed lines are the first zeta ordinates. This is an illustration of the classical explicit-formula modes; the peak-matching statistic is not used inferentially because of selection bias and finite-window resolution.

K. Prime-counting variance comparison. Proposition 10.4 puts the residual on the prime-counting side. At prime scale $X \simeq 4.3 \times 10^6$, the measured variance of short-interval prime counts was compared with the leading Montgomery–Soundararajan-type scale and with the Cramér scale. The reported ratios are

H	200	1000	5000	20000
$\mathrm{Var}_{\mathrm{meas}} / \mathrm{Var}_{\mathrm{MS}}$	1.116	1.202	1.310	1.622
$\mathrm{Var}_{\mathrm{meas}} / \mathrm{Var}_{\mathrm{Cramer}}$	0.552	0.468	0.372	0.314

The decreasing ratio to the Cramér scale is the expected qualitative signature of an $H \log(X/H)$ variance rather than $H \log X$. The excess over the leading MS term is not interpreted as a new statistic: at $H = 2 \times 10^4$ the displayed leading logarithmic factor is only about 2.96, versus $\log X \simeq 15.3$, so lower-order terms are not negligible in this preasymptotic range.

L. Cross-coupling test. The only numerical loophole left inside E_{mid} is weaker than new endpoint information. The gap factor in Theorem 10.1 is controlled by prime statistics, while the window length is $h = |M(n)|$. One can ask whether the standardised fluctuation

$$Z(n) = \frac{E_{\mathrm{mid}}(n)}{\mathrm{SD}_{\mathrm{cov}}(n)}$$

is coupled to the sign or to a zeta phase of M. The supplied follow-up tests reported

test	value
$\operatorname{corr}(Z, \operatorname{sgn} M)$	$+0.00058$
$\operatorname{corr}(Z, M/\sqrt{n})$	$+0.00110$
$\operatorname{corr}(Z, \cos(\gamma_1 \log n))$	-0.00119
$\operatorname{corr}(Z, \sin(\gamma_1 \log n))$	$+0.00173$
$\overline{Z}\,\vert_{M>0} - \overline{Z}\,\vert_{M<0}$	$+0.00027$ $\quad(0.08\sigma$ effective$)$

and an eight-bin phase test gave dispersion of the bin means 0.00070, compared with a typical standard error 0.00499.

These are null results only at the tested resolution. With effective sample $n_{\text{eff}} \simeq 17\,316$, the nominal standard error of a small correlation is $1/\sqrt{n_{\text{eff}}} \simeq 0.0076$, so the experiment excludes only linear couplings of order $|r| \gtrsim 0.015$. Weaker effects, nonlinear dependence, untested functions of the phase, or effects emerging only at much larger n remain outside the sensitivity of this computation. The result therefore does *not* prove decoupling; it says that no coupling was detected in this small specified family.

13. Status of the claims

To keep the register unambiguous:

- *Proved*: Lemmas 2.1, 2.2, 10.2; Propositions 3.3, 4.1, 5.1, 8.1, 9.2; Corollary 4.2; Theorems 6.2, 10.1.
- *Measured*, over $10^4 \le n \le 10^6$ with effective sample $\approx 1.7 \times 10^4$ where relevant: the law $|E_{\text{mid}}| \approx 0.155\,|M|^{3/2} \log n$; the failure of the independent-gap model by a factor 0.853 and its repair by the measured gap covariance; approximate normality of the standardised residual; the 0.9939 packet-envelope correlation; the finite-dimensional recovery of the first lags of ρ from κ; the short-interval variance ratios; and the null cross-coupling tests of Section 12.
- *Classical interpretation, not novelty*: the frequencies γ_k in the $u = \log n$ spectrum of $M(e^u)e^{-u/2}$ are the modes supplied by the explicit formula. The displayed peak matching is diagnostic only and has frequency resolution about 0.91.
- *Conjectural*: that $Z_{3/2}$ converges in distribution; that the limit is Gaussian; that a central limit theorem holds for the antisymmetric weighted gap sum of Theorem 10.1; or that the observed numerical absence of cross-coupling persists asymptotically. None follows from the present numerics.

14. What the construction does and does not deliver

It delivers an exact, elementary dictionary between SS and (M, Q), a two-sided transfer of the Mertens criterion (Theorem 6.2), and a residual whose size obeys a clean measured scaling law.

It does not deliver an independent arithmetic constraint on $M(n)$, by Proposition 8.1. For a transformed representation to become useful it would have to produce at least one of:

(i) a cancellation unavailable in the original variables;
(ii) a positivity or monotonicity principle strong enough to yield new bounds;
(iii) a spectral representation turning a Mertens bound into an operator inequality;
(iv) an inequality between independently controlled components;
(v) a mechanism by which the prime side constrains the Möbius side beyond what (M, Q) already encodes.

None of these has been extracted here. The measured law of Section 12 is of type (iv) in form only: it bounds E_{mid} in terms of $|M|$, which is the wrong direction to be useful. The cross-coupling experiment tests whether the prime-gap fluctuation might nevertheless provide an independently controlled handle on the Mertens side; no such handle is detected, but the numerical sensitivity is only at the 10^{-2} correlation scale and therefore does not close every conceivable nonlinear or asymptotic coupling.

15. A variant that is not an endpoint statistic (speculative)

The obstruction identified in Proposition 8.1 is caused by the use of two *separate* ordinal counters, which destroys the ordering of the ± 1 occurrences. The minimal modification that removes it is a *shared* counter:

$$\widetilde{S}(k) := \mu(k)\, p_{Q(k)}, \qquad \widetilde{SS}(n) := \sum_{k \le n} \mu(k)\, p_{Q(k)}.$$

Here the prime index depends on k rather than on the sign-specific ordinal, so $\widetilde{SS}(n)$ is not a function of $(M(n), Q(n))$: by Abel summation with $w(k) = p_{Q(k)} \asymp k \log k$,

$$\widetilde{SS}(n) = M(n)w(n) - \sum_{k<n} M(k)\big(w(k+1) - w(k)\big),$$

so it depends on the whole trajectory of M and not only on its endpoint. This is offered as a direction, not a result; it is close to the classical weighted Mertens sums $\sum_{k \le n} \mu(k) k^{\alpha}$ and may well reduce to them.

16. Novelty status

The three zeta ingredients are classical. The spectral occurrence of the γ_k in $M(e^u)e^{-u/2}$ is also classical explicit-formula content, and $\kappa(h)$ is exactly a linear functional of the prime-gap autocorrelation sequence, not a new GUE-type statistic. GUE concerns zero spacings; the prime gaps entering Theorem 10.1 belong to a different arithmetic statistical problem.

A search did not locate the sequence S $(2, -2, -3, 1, -5, 3, -7, 1, 1, 5, -11, 1, \dots)$ as a named sequence or standard transform. Absence of a search result is weak evidence and the search was not exhaustive; this is recorded as an observation about the formulation, not a claim about its consequences, which by Proposition 8.1 follow from (M, Q). The genuinely non-endpoint variant in the preceding section remains a direction rather than a result.

17. Summary

$$\mu \longrightarrow (N_-, N_0, N_+) \longrightarrow S \longrightarrow SS, \qquad N_{\pm} = \frac{Q \pm M}{2}, \quad N_0 = n - Q, \quad Q \equiv M \ (2),$$

$$\boxed{SS(n) = \mathcal{P}\left(\frac{Q+M}{2}\right) - \mathcal{P}\left(\frac{Q-M}{2}\right) + n - Q}$$

$$\boxed{T(n) = SS(n) - n + Q(n) = \operatorname{sgn}(M) \sum_{j \in W(n)} p_j, \quad |W(n)| = |M(n)|, \text{ centre } \tfrac{Q(n)+1}{2}}$$

$$\boxed{T(n) = M(n)\, p_{\lfloor Q(n)/2 \rfloor}\big(1 + \theta(n)\big), \quad \theta(n) \to 0}$$

$$\boxed{\text{RH} \iff T(n) = O_{\varepsilon}\big(n^{3/2+\varepsilon}\big) \ \ \forall \varepsilon > 0}$$

$$\boxed{\begin{aligned} T(n) &= M(n)\, p^*(n) + E_{\text{mid}}(n), \\ E_{\text{mid}}(n) &= \operatorname{sgn}(M) \sum_i w_i g_i, \\ \sum_i w_i^2 &= \frac{h^3}{12}\big(1 + O(h^{-1})\big). \end{aligned}}$$

$$\boxed{\text{SD}\big(E_{\text{mid}}\big) \ = \ \sigma_g(n)\sqrt{w^{\top} R\, w}\,, \qquad |E_{\text{mid}}(n)| \approx 0.155\, |M(n)|^{3/2} \log n}$$

The gap representation and its prime-counting integral form are exact; the 0.155 scaling and distributional statements are measured over $10^4 \le n \le 10^6$ and are not proved. The packets are the excursions of $|M|$ amplified by the $h^{3/2}$ weight norm. Their zeta frequencies on $u = \log n$

are classical explicit-formula modes, and the tested standardised gap factor shows no detectable linear coupling to the sign or first zeta phase of M at current sensitivity.

Every quantity above is a function of $(M(n), Q(n))$. The transform is an exact, prime-amplified reparametrisation of the Mertens function; the residual E_{mid} trades the arithmetic of μ for the statistics of prime gaps in a window of length $|M(n)|$, at the cost of a factor $|M(n)|^{3/2}$, and therefore cannot improve a bound on $M(n)$.